

TrES-1b

R-1873

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_130803 · created 2026-08-03T10:59:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456507.7657 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.37 +/- 1.99 [%]
Orbital Inclination (inc)	85.77 +/- 1.45
Airmass coefficient 1 (a1)	0.152 +/- 0.04
Airmass coefficient 2 (a2)	1.66 +/- 0.4
Scatter in the residuals of the lightcurve fit is	22.18 %
Best Comparison Star	#1 - [540, 248]
Optimal Aperture	3.74
Optimal Annulus	6.84
Transit Duration (day)	0.075 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.054 vs 0.1358 (deviation 0.58σ)
Duration fit vs published	0.075 d vs 0.1042 d (deviation 0.67σ)
Mid-time O–C	2.5 min
Depth SNR	2.2
Residual scatter	22.18 %

Light curve

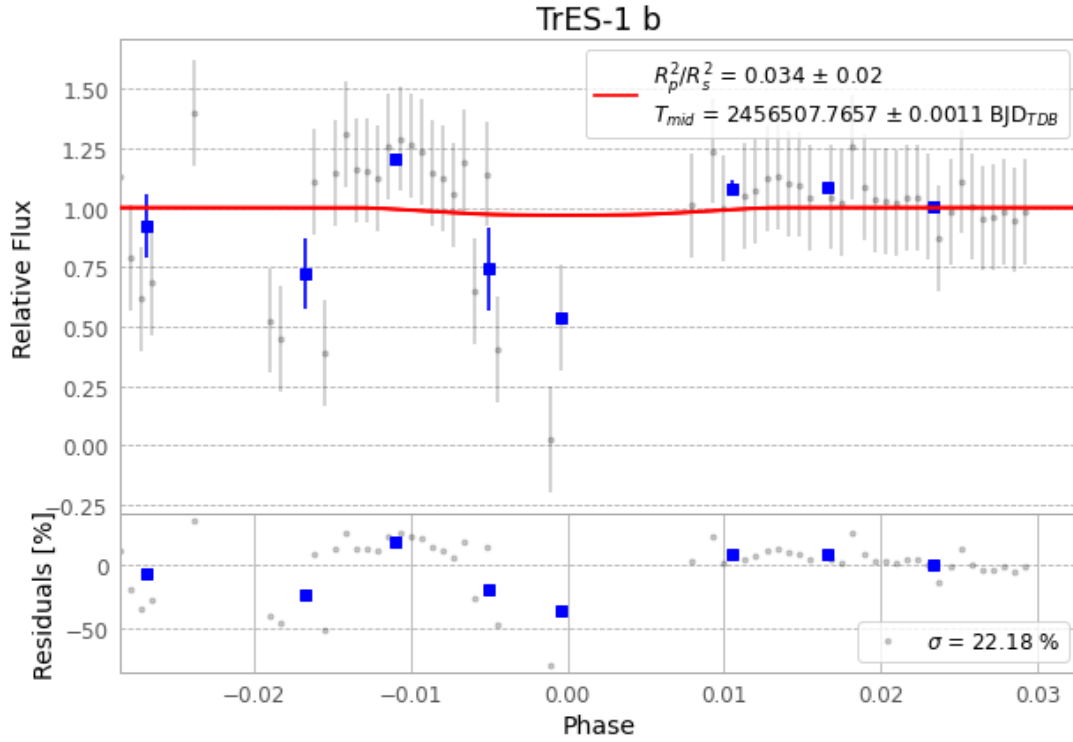


Figure 1 — FinalLightCurve_TrES-1 b_2013-08-02.png (SHA-256 50d7655c2e51cb94...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 270], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 42661fe25f8c895...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), TRES-1130803041212.FITS → TRES-1130803083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-130803.log (cb131e699e91...), FinalLightCurve_TrES-1 b_2013-08-02.png (50d7655c2e51...), FinalLightCurve_TrES-1 b_2013-08-02.csv (f22524fe3c4f...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 10:59Z.